Biochemistry Cume

Protein Chemistry

Max. Points: 100 points

1. Name three chemical reagents which are popularly used for the determination of N-terminal amino acid of proteins/peptides. (**9 points**).

2. The polypeptide chain given below is treated independently with cyanogen bromide and trypsin. The amino acid or amino acid sequences of the fragments are listed below in their single letter codes. Is there enough information to derive the complete amino acid sequence of the polypeptide chain? If so, what is the amino acid sequence of the entire polypeptide? (**26 points**)

1. Cyanogen bromide treatment

a. K
b. R-F-M
c. M-L-Y-C-R-G-M
d. N-I-K-G-L-M

2. Trypsin Hydrolysis

e. F-M-K
f. G-M-N-I-K
g. G-L-M-R
h. M-L-Y-C-R

3. Predict the products formed when the polypeptide, **G-R-T-M-Y-K-S-P-W-M-F-K-M**, is treated with cyanogen bromide. Assume the reaction of the polypeptide with cyanogen bromide is complete. (**15 points**).

4. A bioactive peptide is isolated from the bacterium, *Bacillus brevis*. The structure of the peptide is deduced from the following observations. (**50 points**)

A. Complete acid hydrolysis of the bioactive peptide followed by amino acid analysis yields equimolar amounts of Leu, Lys, Phe, Pro, and Val.

B. The molecular weight of the peptide is 1,200 Daltons.

C. The peptide is resistant to carboxypeptidase treatment. (Carboxypeptidase is an enzyme which catalyzes the release of the C-terminal most amino acid of a polypeptide unless that residue is Pro or does not contain a free α-carboxyl group for some reason.)

D. Treatment of the intact bioactive peptide with 1-fluoro-2,4-dintrobenzene (FDNB), followed by complete hydrolysis and chromatography, yields only free amino acids and DNB-ε-NH-Lys.

E. Partial acid hydrolysis of the bioactive peptide followed by chromatographic separation and sequence analysis yielded the following di- and tripeptides:

Leu-Phe, Phe-Pro, Lys-Leu, Val-Lys, Val-Lys-Leu, Phe-Pro-Val, Pro-Val-Lys

Given the above information, deduce the amino acid sequence of the bioactive peptide. Assume that the average molecular weight of an amino acid as 110 Daltons.